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Botanical mitigation of Herpes Simplex Virus: A comprehensive review

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Abstract- Herpes simplex virus type 1 (HSV-1) and type-2 (HSV-2) are extensively prevalent, infectious human pathogens causing oral and genital lesions as primary infections. They establish a latent neural infection reservoir in neural ganglion and occasionally there is recurrence in response to various physiological stimulants after primary infection. Additionally, associated infection includes herpes keratitis, encephalitis, blindness, and genital tract inflammation. Acyclovir, Famciclovir, Valacyclovir and their derivatives are generally used for the treatment of HSV but the development of resistance and side effects limit potential use of drugs against it. Plants have bewildering nature of prophylaxis agents which can used to harnessed to treat various disease ailments. Botanicals i.e. plant's essential oil, extracts and isolated compounds have antiviral activity against HSV with the diverse mode of action to treat the herpetic infections. This review aims to summarise the current status of knowledge of the use of botanicals as antiviral agent against Herpes Simplex Virus.

Key words: Herpes Simplex Virus, Botanicals, Antiviral, Prophylaxis, Antiviral Mechanism.

INTRODUCTION

Viral diseases uncourtly are one of the majors engenders of mortality worldwide. Some of the deadliest viral infections are SARS (Severe Acute Respiratory Syndrome), influenza, Ebola and AIDS.[1,2] Herpes simplex viruses (HSV) is a highly infectious virus with a double stranded linear DNA, occurring frequently causing labial oral genital lesion in human, without exhibiting seasonal changes and affect people of all age groups.[3,4] WHO estimated 3.7 billion people under age 50 have 67% HSV-1 and 491 million between 15-49 have 13% HSV-2 infection worldwide.[5] HSV belongs to family Herpesviridae. The family contains a total 8 members. Family comprises 3 sub families based on biological and genomic analysis characteristics. These are α, β and γ. Sub family α includes HSV-1, HSV-2 and Varicella Zoster Virus (VZV). HSV-1 & HSV-2 show approx. 78% similarity of amino acid.[6] Both HSV-1 & HSV-2 causes lifelong infections. Primary infection site of HSV-1 is oral and perioral region i.e. mouth, nose, lips, eyes etc. Serious complications of HSV-1 infections cause wound around orolabial regions, neonatal infections, keratitis, meningitis and encephalitis.[7,8] HSV-2 mostly causes genital herpes infections.[9] In due course of infection virus establishes itself in a dormant latency in trigeminal or lumbosacral ganglion (sensory ganglion) and persist in CNS lifelong. The fundamental biological feature of HSV are neurotropic invasion and destruction, neurotoxicity and latent

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dormancy in neural dendrites of sensory ganglia.[10] Humans are natural host of HSV. But under immunosuppression, trauma, stress, ultraviolet radiation, hormonal disbalance, bacterial infection of respiratory tract, virus reactivates, and clinical symptoms can be exhibited or it can be asymptomatic. Asymptomatically transmission of HSV occur by shedding. Mainly HSV transmitted through genital surface, oral to oral and oral to genital region. HSV can be directly or indirectly transmitted from one to another individual through lesions and form its fluid secretion.[11]

Diagnostics of HSV

Diagnosis of HSV-1 and HSV-2 at first based on presence of lesion or ulcer. Viral culture primarily is a gold standard diagnostic method and takes 7 days to confirmation. Time taken during viral culture diagnosis can worsen the patients' symptoms, spread of infection and also needs 48 hours sample otherwise it lowers the sensitivity. Antigens specific direct fluorescent antibody testing, serology, HSV PCR and Multiple HSV PCR techniques are available for the diagnosis and detection of viral load[12-14], HSV PCR are more sensitive method than the viral culture detection.[15] Miyachi *et al.* (2022)[16] diagnoses HSV from perioral and genital areas by using loop-mediated isothermal amplification (LAMP) method and amplifies target DNA sequences. LAMP is a simpler, faster and reliable method that can differentiate primary, recurrent and other conditions. Natarajan *et al.* (2022)[17] revealed that infectious keratitis and viral keratitis can be diagnosed by artificial intelligence-based deep learning (DL) algorithms. Diagnosis of infectious keratitis is very important because misdiagnosis leads to ophthalmic morbidity. Metagenomic next-generation sequencing is one of the robust techniques extensively used for the detection and diagnosis of herpes simplex encephalitis from cerebrospinal fluid. This technique is faster and more reliable than conventional culture methods and PCR.[18] Figure 2 showing the infection regions of HSV on human.

Management of HSV and Treatment

Antiviral drugs Acyclovir are the first line of treatment given during the HSV infections to cure the lesion and to suppress the symptoms. Famciclovir, Valacyclovir, Foscarnet, Cidofovir, Ganciclovir, Penciclovir also available and are effective prophylactics against HSV infections. These drugs are nucleic acid analogs which interferes and disrupts the functions of DNA polymerase.[19] These drugs reduce lesions and frequency of recurrence but do not cure the herpes completely. Permanently cure of HSV in present days in not available.[20]

Natural Therapy

Plants generate an array of millions of secondary metabolites and phytochemicals, secreted in low or high quantity and performs defensive role in plant physiology. Many of these phytochemicals in micron or submicron act as antimicrobials. According to WHO reports 65-80% populations use plant-based product as therapeutics. Plant kingdom have potent medicinal compounds to treat the infectious diseases.[21] Plant based essential oil, extracts, isolated chemical compound play a central role as therapeutics and prophylaxis agent athwart to infectious viral diseases. Essential oils are complex odoriferous mixtures of various active constituents i.e. terpenes, terpenoids, hydrocarbons, ethers, alcohols, carotenoids, ketones, aldehydes, aromatic phenols and phenylpropene derivatives.[22,23] Substantially plant extract also owns diverse composition natural occurring compounds i.e. phenols, polyphenols, saponins, alkaloids, glycosides and flavonoids.[24-26] They have diverse phytochemicals with diverse mechanism of action on viral agents at various stages. Exploitation of botanicals from plants can be a safe and promising to combat infectious and resistant HSV in modern era.[27] Rigorously use of present anti-Herpes medication, like acyclovir have led to the emergence of resistant HSV infection.[28] Álvarez *et al.* (2020)[29], Garber *et al.* (2021)[30] and van *et al.* (2021)[31] summarized botanical, plant extract and active compounds against herpes simplex virus. Use of synthetic chemicals as drugs has been restricted due to their cytotoxicity and environmental adverse effect. Table 1, 2 and 3 are report use of Plant extract, Essential oils and plant derived chemicals against HSV-1 & HSV-2.

Table 1. Efficacy of Plant Extract against HSV-1 & HSV-2.

Sl.No.	Plant	Virus	Cell Line Used	Dosage	Reference
1.	*Scaevola sericea* Forst (Leaf)	HSV-2	Vero	62.5(µg/mL)	32
2.	*Psychotria hawaiiensis Gray* (Leaf)	HSV-2	Vero	250(µg/mL)	
3.	*Pipturus albidus Gray* (Stem)	HSV-2	Vero	250(µg/mL)	
4.	*Eugenia malaccensis* L. (Bark)	HSV-2	Vero	125(µg/mL)	
5.	*Helichrysum aureonitens* Sch. Bip.	HSV-1	Human Lung fibroblast	1.35 mg/ml	33

6.	*Trixis praestans* (Vell.) Cabr.	HSV-1 & HSV-2	Vero	214 &250 (µg/mL)	34
7.	*Cunila spicata* Benth. (Aerial part)	HSV-1 & HSV-2	Vero	70 (µg/mL)	
8.	*Satureja boliviana* (Benth.) Briquet	HSV-1	HeLA	10 to 125 (µg/mL)	35
9.	*Baccharis genistelloides* (Lam.) Pers	HSV-1	HeLA	25 to 50 (µg/mL)	
10.	*Galanthus elwesii L. (bulb)*	HSV	Vero	8(µg/mL)	36
11.	*Rheum ribes L. (rhizome)*	HSV	Vero	8(µg/mL)	
12.	*Astragalus membranaceus*	HSV-1	2BS cell	0.98(g/mL)	37
13.	*Carissa edulis (Forssk.) Vahl (root)*	HSV-1	Vero	50(µg/mL)	38
14.	*Eugenia caryophyllus* (Spreng.) Bullock & S. G. Harrison (Flower bud)	HSV-1 & HSV-2	GMK	50.2-72.8(µg/mL)	39
15.	*Pelargonium sidoides DC (root)*	HSV-1 & HSV-2	RC-37	-------	40
16.	*Tripterygium hypoglaucum* (Level) Hutch	HSV-1	Vero	6.5(µg/mL)	41
17.	*Cleome rosea* (Vahl) (Leaves)	HSV-1 & HSV-2	Vero	12.5(µg/mL)	42
18.	*Dianthus caryophyllus* L. (Seed)	HSV-1	Vero	20 µg	43
19.	*Melissa officinalis* L.	HSV-1	RC-37	15 (µg/mL)	44
20.	*Echium amoenum* L.	HSV-1	Hep 2	400 (µg/mL)	45
21.	*Phyllanthus watsonii* (A. Gray) J.M. Porter & L.A. Johnson	HSV-1 & HSV-2	Vero	11.9 ± 6.8 (µg/mL)	46
22.	*Phyllanthus urinaria* L.	HSV-1 & HSV-2	Vero	<12.5 (µg/mL)	46
23.	*Schinus terebinthifolius* Radii	HSV-1	Vero	10.2-15.51 (µg/mL)	47
24.	*Moringa oleifera* Lam. (leaves)	HSV-1 & HSV-2	Vero	200 (µg/mL)	48
25.	*Veronica persica* Poir	HSV-1	Vero	120 (µg/mL)	49
26.	*Equisetum giganteum* L.	HSV-2	Mouse model	18 (µg/mL)	50
27.	*Vachellia nilotica* (L.)	HSV-2	Vero	4.71 (µg/mL)	51
28.	*Graptopetalum paraguayense* E. Walther	HSV-1 & HSV-2	Vero	0.0001 mg/mL	52
29.	*Tanacetum parthenium* (L.) Sch. Bip.	HSV-1	Mice	4 -8(mg/kg)	53
30.	*Sarracenia purpurea* L.	HSV-1	Vero	23 (µg/mL)	54
31.	*Macrocystis pyrifera*	HSV-1 & HSV-2	Vero	0.07 mg/mL	55
32.	*Durvillaea antarctica*	HSV-1 & HSV-2	Vero	0.08 mg/mL	
33.	*Peganum harmala* (L.) (seed)	HSV-2	Vero	3.63±0.58 (µg/mL)	56
34.	*Pistacia lentiscus* L. (stem)	HSV-2	Vero	161±12.12 (µg/mL)	
35.	*Stephania hernandifolia* (Willd.) Walp.	HSV-1	Vero	4.32 -4.50 (µg/mL)	57

Table 2. Effective Essential oils against HSV-1 & HSV-2.

Sl.No.	Plant	Virus	Cell Line Used	Dosage	Reference
1.	*Salvia fruticosa* Miller	HSV-1	Vero	---	58
2.	*Santolina insularis*	HSV-1 & HSV-2	Vero	0.88 & 0.7 (µg/mL)	59
3.	*Australian tea* Tree	HSV-1 & HSV-2	RC-37	0.0009% &0.0008%	60
4.	*Eucalyptus*	HSV-1 & HSV-2	RC-37	0.009% & 0.008%	
5.	*Mentha piperita* L.	HSV-1 & HSV-2	RC-37	0.002% & 0.0008%	61
6.	*Aloysia gratissima* (Gillies & Hook.)	HSV-1	Vero	65 ppm	62
7.	*Artemisia douglasiana Bess.*	HSV-1	Vero	83 ppm	62
8.	*Eupatorium patens* (D. Don ex Hook. & Arn.	HSV-1	Vero	125 ppm	
9.	*Tessaria absinthioides* (Hook & Arn.) DC.	HSV-1	Vero	105 ppm	
10.	*Melaleuca armillaris* (Smith)	HSV-1	Green monkey kidney cell	-------	63
11.	*Melissa officinalis* L.	HSV-2	Hep-2	20-200 (µg/mL)	64
12.	*Leptospermum scoparium* J.R.Forst. et G.Forst	HSV-1 & HSV-2	RC-37	0.96 &0.58(µg/mL)	65
13.	*Eugenia caryophyllus* (Spreng.) Bullock & S. G. Harrison	HSV-1	GMK	-----	66
14.	*Illicium verum*	HSV-2	RC-37	0.016%(IC50)	67
15.	*Hyssopus officinalis*	HSV-2	RC-37	0.0075(IC50)	
16.	*Thymus vulgaris*	HSV-2	RC-37	0.007%(IC50)	
17.	*Zingiber officinale*	HSV-2	RC-37	0.004%(IC50)	
18.	*Matricaria recutita*	HSV-2	RC-37	0.003%(IC50)	
19.	*Santalum album*	HSV-2	-----	0.0015%(IC50)	
20.	*Melissa officinalis L.*	HSV-1 & HSV-2	monkey kidney cell	0.0004% and 0.00008%(IC50)	68

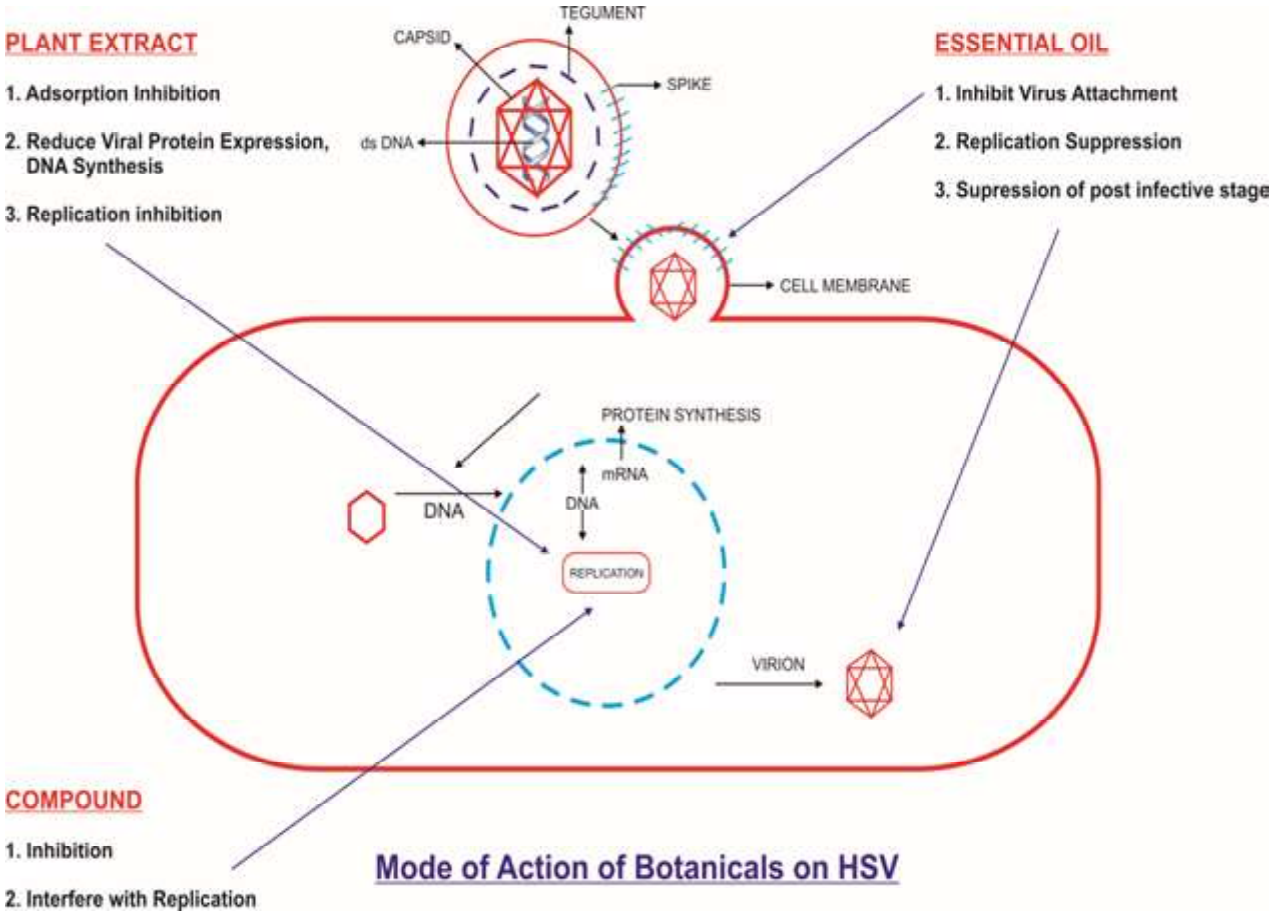
21.	*Lantana grisebachii* Seckt.	HSV-1 & HSV-2	Vero	26.1 ppm	69
22.	*Glechon spathulate* Benth.	HSV-1	Vero	0.0095%	70
23.	*Glechon marifolia* Benth.	HSV-1	Vero	0.039%	
24.	*Zataria multiflora* Boiss	HSV-1	Vero	0.003% (IC50)	71
25.	*Eucalyptus caesia*	HSV-1	Vero	0.004%(IC50)	
26.	*Artemisia kermanensis*	HSV-1	Vero	0.007%(IC50)	
27.	*Satureja hortensis* L.	HSV-1	Vero	0.008%(IC50)	
28.	*Rosmarinus officinalis*	HSV-1	Vero	0.006%(IC50)	
29.	*Hyptis mutabilis*	HHV-1 & HHV-2	Vero	50(µg/mL)	72
30.	*Lepechinia vulcanicola*	HHV-1 & HHV-2	Vero	100(µg/mL)	
31.	*Minthostachys mollis*	HHV-1 & HHV-2		100(µg/mL)	
32.	*Ocimum campechianum*	HHV-1 & HHV-2	Vero	100 & 50(µg/mL)	
33.	*Melaleuca alternifolia*	HSV-1	Vero	10% v/v	73
34.	*Rosa damascene* Mill.	HSV-1	MDBK	--	74
35.	*Rosa alba* L.	HSV-1	MDBK	--	
36.	*Phlomis aurea* Decne.	HSV-1	------	25	75
37.	*Acacia nilotica* (L.)	HSV-1 & HSV-2	------	14.26 ± 0.54 and 3.99 ± 0.15 (%)	76
38.	*Santolina impressa* Hoffmanns. and Link	HSV-1 &HSV-2	------	2.54 µg/mL and 1.59 µg/mL	77

Table 3. List of different plant derived compounds against HSV-1 & HSV-2.

Sl. No.	Plant	Derived compound	Virus	Cell Line used	Dosage	References
1.	*Crataegus sinaica* Boiss	(a) o-glycosidic flavonoids, (b) Proanthocyanidin	HSV-1	Vero	----	78
2.	*Melia azedarach* L.	Meliacine	HSV-1	Vero	50 µg/mL	79
3.	*Leptospermum scoparium* J.R.Forst. et G.Forst	Flavesone, Leptospermone	HSV-1 & HSV-2	RC-37	----	80
4.	*Cassia javanica*	ent-epiafzelechin-(4α→8)-epiafzelechin	HSV-2	Vero	166.8±12.9 µM	81
5.	*Eugenia caryophyllus*	Eugenol	HSV-1	GMK	149.5 & 107.1 µg/mL	82
6.	*Artocarpus lacucha*	Oxyreservatrol	HSV-1	Mouse mode	500 mg/kg/dose	83
7.	*Capitidis rhizome*	Berberins	HSV-1 & HSV-2	Vero	8.2 ± 1.2 9 x 10^{-2} & 9.0 ± 1.1 9 x 10^{-2} mg/ml	84
8.	*Phyllanthus orbicularis*	I. epicatechin-3-O-gallate	HSV-2	Vero	11.7 mg/mL	85
		II. Procyanidins B1 and B2	HSV-2	Vero	32.8 mg/mL & 24.2 mg/mL	
9.	*Ferula assafoetida*	Badrakemin	HSV-1	Vero	10, 5& 2.5 µg/mL	86
10.	*Azadirachta indica*	Sulfonoquinovosyldiacylglyceride	HSV-1 & HSV-2	Vero	9.1 & 8.5 µg/mL	87
11.	*Astonia scholaris*	17-nor-excelsinidine, strictamine	HSV	Vero	1.09 µg/mL,0.36 µg/mL	88
12.	*Peganum harmala*	Harmine	HSV-1 & HSV-2	Hec-1A cells	1.47 Mm	89
13.	*Camellia sinensis*	I. Theaflavin	HSV-1	Vero	50 µM	90
		II. Theaflavin-3-monogallate	HSV-1	Vero	25 µM	
		III. Theaflavin-3,3'digallate	HSV-1	Vero	20 µM	
14.	*Kalanchoe daigremontiana*	I. Kaempferol 3-O-β-d-xylopyranosyl-(1 → 2)-α-l-rhamnopyranoside	HSV-1 & HSV-2	Vero	7.4 & 9.0 µg/mL	91
		II. quercetin 3-O-β-d-xylopyranosyl -(1 → 2)-α-l-rhamnopyranoside	HSV-1 & HSV-2	Vero	5.8 & 36.2 µg/mL	
15.	*Houttuynia cordata*	Houttuynoid A	HSV-1 & HSV-2	Vero	36.38±6.01µM	92
16.	*Fridericia formosa*	I. 2′-*O-trans*-coumaroylmangiferin	HSV-1	Vero	47.4 ± 6.1 µg/mL	93
		II. 2′-*O-Trans*-cinnamoylmangiferin	HSV-1	Vero	77.4 ± 4.3 µg/mL	
17.	*Eucalyptus globulus*	I. Tereticornate A	HSV-1 & HSV-2	Vero	0.96 µg/mL	94
		II. Cypellocarpin C	HSV-1 & HSV-2	Vero	0.73 ± 0.11 µg/mL	
		III. Grandinol	HSV-1 & HSV-2	Vero	1.23 ± 0.13 µg/mL	
		IV. Sideroxylin	HSV-1 & HSV-2	Vero	1.44 ± 0.14 µg/mL	
18.	*Ampelopsis grossedentata*	Dihydromyricetin	HSV-1	Vero	12.56 µM	95
19.	*Peganum harmala*	I. Harmine	HSV-2	Vero	4.06±0.68 µM	96
		II. Pegaharine D	HSV-2	Vero	2.12 ± 0.14 µM	
20.	*Erythrina speciosa*	Vitexin	HSV-1	Vero	18 ± 3.3 µg/mL	97
21.	*Jatropha multifida*	15-epi-4E-jatrogrossidentadion	HSV-1	Vero	2.05 µg/mL	98

Mode of action of botanicals as antiviral

Natural products from plants have diverse mode of action on HSV i.e. inhibition of virus adhesion, attachment, replication and post replication processes. *Santolina insularis* essential oils inactivates and prevent the virion adsorption to the host cell.[59] *Salvia desoleana* essential oil interfere with replication and suppress replication after virus entry.[99] Extract of *Peganum harmala* and *Pistacia lentiscus* are active against HSV-1 and HSV-2 act as virucidal after adsorption and penetration in cell.[56] *Nepeta nuda* extract exhibited antiviral activity preventing adsorption and influencing replication against HHV (Human Alpha Herpes Virus, Genus- Simplex Virus).[100] Not only the essential oil or plant extract but the isolated compounds from plants can be the potent one on HSV infections. Flavonoid Dihydromyriectin (DHM) isolated from *Ampelopsis grossedentata* declines early and late gene expression in HSV-1[95] and Ent-epiafzelechin-(4α→8)-epiafzelechin (EEE) extracted from *Cassia javanica* leaves inhibit HSV-2 replication.[81] Figure 1 shows several modes of action of botanicals against HSV. Fig. 1



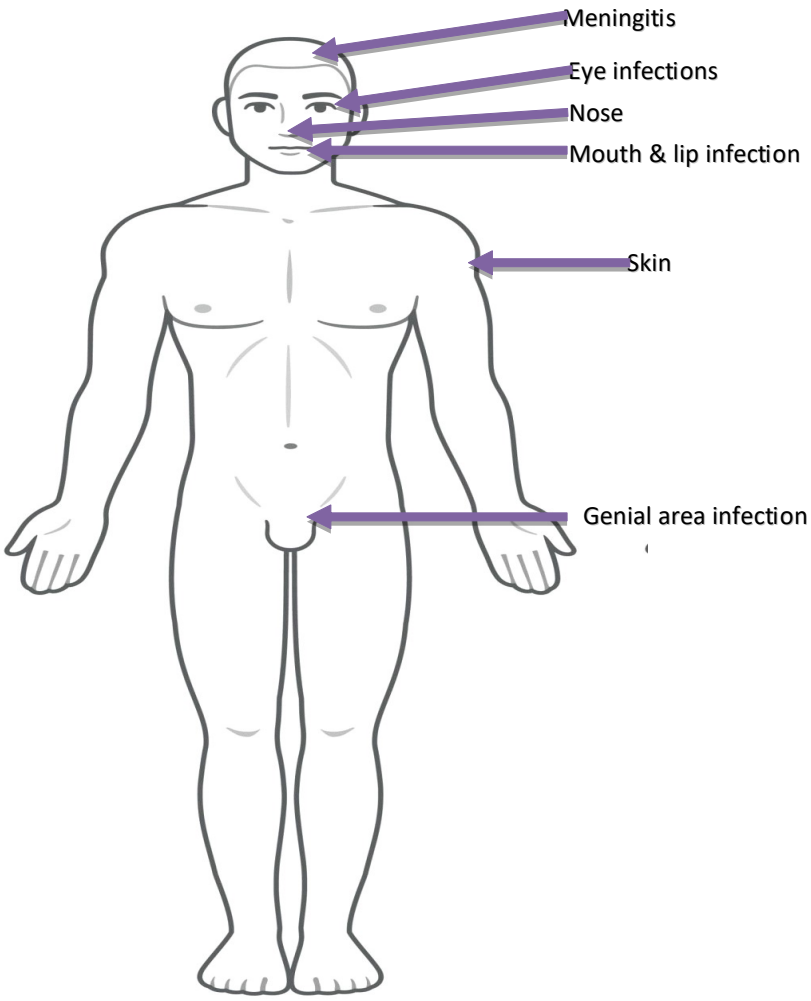


Fig 2. Showing the HSV-1 and HSV-2 infection sites on human body.

CONCLUSION

Herpes simplex virus (HSV) is ubiquitously prevalent in human population worldwide. Recurrence and neurotropism of HSV-1 and HSV-2 is a global health issue on the landscape. Conventional therapy increases the risk of resistance to common antiviral drugs presently available. There is urgent need of alternative therapy to counter the HSV infections. Presently we reviewed work done on plant extracts, essential oil and their isolated compounds, which have anti-HSV activity having inhibiting adsorption, penetration, DNA replication and post infective stages. These nominee botanicals might be the alternative treatment for HSV resistant virus. There is much research needed to investigate the antiviral activity of resistant clinical isolates of HSV-1 and HSV-2. Although conventional drugs directly or indirectly inhibit replication via acting as a nucleoside analogue but the survival of HSV is noticeable. Distinctly there is a need of analysis of botanicals to evaluate and investigate anti- HSV properties. Plant extract, essential oil and the purified isolated compounds have been investigated for their antiviral effectiveness against HSV-1 and HSV-2. Extended research is needed to explore more effective plant-based compound which can be commercialized in future as therapy of HSV. Issues like mode of action, sustainability, mammalian toxicity, development of drug resistance must be addressed along the way. With the advent of modern tools in drug designing and synthetic chemistry plant provide a sea of opportunities to explore

diverse compounds of interest which can be harnessed as antiviral drugs for human welfare in times to come.

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